

# Write Code with using given Datatypes

List, Tuple, Set, Dictionary and String, with one simple example each.

## 1. List

**What is it:** A list stores many values in order, inside square brackets [ ].

**Why we need it:** It lets us keep related items together and change them later.

**Where it is used:** Used for storing marks, names, shopping items, or any changing group of data.

### Simple code

```
| marks = [80, 70, 95]  
| marks.append(60)  
| print(marks)
```

*Output: [80, 70, 95, 60]*

## 2. Tuple

**What is it:** A tuple is like a list but written with round brackets ( ), and it cannot be changed after it is made.

**Why we need it:** It keeps data safe from being changed by mistake.

**Where it is used:** Used for fixed data, like coordinates (x, y) or a date (day, month, year).

### Simple code

```
| point = (2, 3)  
| x, y = point  
| print(x, y)
```

*Output: 2 3*

## 3. Set

**What is it:** A set stores values with no order and no repeats, written with curly brackets { }.

**Why we need it:** It removes duplicate values automatically and checks membership fast.

**Where it is used:** Used to find unique items, like unique visitors or unique words in a text.

## Simple code

```
nums = [1, 2, 2, 3, 3, 3]
unique = set(nums)
print(unique)
```

*Output: {1, 2, 3}*

## 4. Dictionary

**What is it:** A dictionary stores data as key and value pairs, written as {key: value}.

**Why we need it:** It lets us find a value quickly using its key, instead of searching one by one.

**Where it is used:** Used for storing details like a student's name and marks, or settings in an app.

### code

```
student = {"name": "Alice", "marks": 90}
print(student["name"])
```

*Output: Alice*

## 5. String

**What is it:** A string is a sequence of characters, written inside quotes.

**Why we need it:** It lets us store and work with text, like names or sentences.

**Where it is used:** Used for text data, such as reading input, printing messages, and processing words.

### code

```
text = "hello"
print(text.upper())
```

*Output: HELLO*

---

## Quick Comparison

| Data Structure | Brackets | Ordered | Can Change | Allows Repeats        |
|----------------|----------|---------|------------|-----------------------|
| List           | [ ]      | Yes     | Yes        | Yes                   |
| Tuple          | ( )      | Yes     | No         | Yes                   |
| Set            | { }      | No      | Yes        | No                    |
| Dictionary     | {k:v}    | Yes     | Yes        | Keys: No, Values: Yes |
| String         | " "      | Yes     | No         | Yes                   |